Mathematical problems

Problem-1: *What is the length of a meter stick moving parallel to its length when its mass is 3/2 of its rest mass?*

Solution: We know

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\frac{m_0}{m} = \sqrt{1 - \frac{v^2}{c^2}} \qquad (1)$$

and

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}} \qquad (2)$$

Here,

$L_0 = 1\ m$

$\frac{m_0}{m} = \frac{2}{3}$

$L = ?$

From equation (1) and (2) we can write

$$L = L_0 \left(\frac{m_0}{m}\right)$$

Or $$L = 1 \times \frac{2}{3}$$

$\therefore$ $$L = 0.667\ m\ (\boldsymbol{Ans})$$

Problem-2: *A spacecrafts is moving relative to the earth. An observer on the earth finds that, according to her clock, 3601 s elapse between 1 p.m. and 2 p.m., on the spacecraft's clock. What is the spacecrafts speed relative to the earth?*

Solution: We know

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$1 - \frac{v^2}{c^2} = \left(\frac{t_0}{t}\right)^2$$

Or $$v = c\sqrt{1 - \left(\frac{t_0}{t}\right)^2}$$

Or $$v = 3 \times 10^8 \times \sqrt{1 - \left(\frac{3600}{3601}\right)^2}$$

$\therefore$ $$v = 7.07 \times 10^6\ ms^{-1} (\boldsymbol{Ans})$$

Here,

$t_0 = 3600\ s$

$t = 3601\ s$

$c = 3 \times 10^8\ ms^{-1}$

$v = ?$

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Problem-3: *The rest mass of a proton is 2000 times the rest mass of an electron. Calculate the speed at which the electron can move.*

Solution: We know

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$\frac{m}{m_0} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$1 - \frac{v^2}{c^2} = \frac{1}{\left(\frac{m}{m_0}\right)^2}$$

Or $$1 - \frac{v^2}{c^2} = \frac{1}{(2000)^2}$$

Or $$\frac{v^2}{c^2} = 1 - \frac{1}{4 \times 10^6}$$

Or $$v = c \times \sqrt{1 - \frac{1}{4 \times 10^6}}$$

Or $$v = 3 \times 10^8 \times \sqrt{1 - \frac{1}{4 \times 10^6}}$$

∴ $$v = 2.99 \times 10^8\ ms^{-1} (\boldsymbol{Ans})$$

Here,

$\frac{m}{m_0} = 2000$

$c = 3 \times 10^8\ ms^{-1}$

$v = ?$

Problem-4: *A particle of mass 10^{-24} kg is moving with a speed of 1.8×10^8 m/s. Calculate its mass when it is in motion.*

Solution: We know

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$m = \frac{1 \times 10^{-24}}{\sqrt{1 - \frac{(1.8 \times 10^8)^2}{(3 \times 10^8)^2}}}$$

∴ $$m = 1.25 \times 10^{-24} kg\ (\boldsymbol{Ans})$$

Here,

$m_0 = 1 \times 10^{-24} kg$

$v = 1.8 \times 10^8\ ms^{-1}$

$c = 3 \times 10^8\ ms^{-1}$

$m = ?$

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Problem-5: *The total energy of a particle is exactly twice its rest mass energy. Calculate its speed.*

Solution: We know

Here,

$mc^2 = 2m_0c^2$

$\frac{m}{m_0} = 2$

$c = 3 \times 10^8 \ ms^{-1}$

$v = ?$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$\frac{m}{m_0} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$1 - \frac{v^2}{c^2} = \frac{1}{(2)^2}$$

Or $$\frac{v^2}{c^2} = 1 - 0.25$$

Or $$v = c \times \sqrt{0.75}$$

$\therefore$ $$v = 2.25 \times 10^8 \ ms^{-1} (\boldsymbol{Ans})$$

Problem-6: *An electron has a kinetic energy of 5.00 keV. Find its speed and mass in terms of its rest mass.*

Solution: We know,

Here,

$K.E. = 5.00 \ keV$

$= 5.00 \times 10^3 \ eV$

$= (5.00 \times 10^3 \times 1.6 \times 10^{-19}) \ J$

$= 8 \times 10^{-16} \ J$

$c = 3 \times 10^8 \ ms^{-1}$

$m_0 = 9.1 \times 10^{-31} kg$

$v = ?$

$m = ?$

$$K.E. = \frac{1}{2} m_0 v^2$$

Or $$v = \sqrt{\frac{2 \times K.E.}{m_0}}$$

Or $$v = \sqrt{\frac{2 \times 8 \times 10^{-16}}{9.1 \times 10^{-31}}}$$

$\therefore$ $$v = 4.19 \times 10^7 \ ms^{-1} (\boldsymbol{Ans})$$

Again

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$m = \frac{9.1 \times 10^{-31}}{\sqrt{1 - \left(\frac{4.19 \times 10^7}{3 \times 10^8}\right)^2}}$$

$\therefore$ $$m = 9.19 \times 10^{-31} kg \ (\boldsymbol{Ans})$$

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Problem-7: *What must be the speed of particle of rest mass* $\frac{1}{2}MeV/c^2$ *be so that its relativistic mass is* $\frac{1}{\sqrt{2}}MeV/c^2$

Solution: We know

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

Or $$\left(\frac{m_0}{m}\right)^2 = 1-\frac{v^2}{c^2}$$

Or $$v^2 = c^2\left[1-\left(\frac{m_0}{m}\right)^2\right]$$

Or $$v = c \times \left[1-\left(\frac{m_0}{m}\right)^2\right]^{1/2}$$

Or $$v = 3\times10^8 \times \left[1-\left(\frac{1}{\sqrt{2}}\right)^2\right]^{1/2}$$

∴ $$v = 2.12\times10^8\ ms^{-1}(\boldsymbol{Ans})$$

Here,

$m_0 = \frac{1}{2}MeV/c^2$

$m = \frac{1}{\sqrt{2}}MeV/c^2$

$\frac{m_0}{m} = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$

$c = 3\times10^8\ ms^{-1}$

$v = ?$

Problem-8: *A man has a mass of 100 kg on the ground. When he is in a spacecraft in flight, his mass is 101 kg as determined by an observer on the ground. What is the speed of the craft?*

Solution: We know

$$m = \frac{m_0}{\sqrt{1-\frac{v^2}{c^2}}}$$

Or $$1-\frac{v^2}{c^2} = \left(\frac{m_0}{m}\right)^2$$

Or $$v = c\sqrt{1-\left(\frac{m_0}{m}\right)^2}$$

Or $$v = 3\times10^8 \times \sqrt{1-\left(\frac{100}{101}\right)^2}$$

∴ $$v = 4.2\times10^7\ ms^{-1}(\boldsymbol{Ans})$$

Here,

$m_0 = 100\ kg$

$m = 101\ kg$

$c = 3\times10^8\ ms^{-1}$

$v = ?$

Problem-9: *A particle is moving with a speed of 0.5c. Calculate the ratio of its rest mass and the mass while in motion.*

Solution: We know

Here,

$v = 0.5\,c$

$\therefore \ \frac{v}{c} = 0.5$

$m = ?$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$\frac{m_0}{m} = \sqrt{1 - \frac{v^2}{c^2}}$$

Or $$\frac{m_0}{m} = \sqrt{1 - (0.5)^2}$$

$\therefore$ $$\frac{m_0}{m} = 0.866\ (\boldsymbol{Ans})$$

Problem-10: *Find the mass of an electron*$(m_0 = 9.1 \times 10^{-31} kg)$ *whose velocity is 0.99c.*

Solution: We know

Here,

$m_0 = 9.1 \times 10^{-31} kg$

$v = 0.99c$

$m = ?$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Or $$m = \frac{9.1 \times 10^{-31}}{\sqrt{1 - \left(\frac{0.99c}{c}\right)^2}}$$

$\therefore$ $$m = 6.45 \times 10^{-30} kg\ (\boldsymbol{Ans})$$

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